Why Domain Matters, part 3

Payments

The record says

`Payment failed.`

What actually happened

The customer’s account was debited.

Saturday 20 June, 6:48 pm. Store 014, Jayanagar.

The till said the payment failed. Her bank had already taken the money.

```
18:48:25 UPI ₹1,064.47 failed: no response
18:48:55 card ₹1,064.47 approved
```

87 seconds after she approved the UPI payment, the store was credited. Nobody at the till saw it. She had paid twice.

Five readings of one evening

Who	What they see	What they conclude
Cashier	“Failed: no response” 28 times in 90 minutes; another payment taken 30 times	“Take another payment.”
Payments operations	Approvals at 69% and 67% for two hours, normal by eight	“Incident closed.”
Reconciliation analyst	17 UPI credits with no approved payment at the till	“Park them in suspense.”
Customer support	26 complaints in June, 23 of them “charged twice”	“Send a screenshot; five days.”
Finance controller	UPI fees of ₹0 in June; ₹11,302 of refunds	“UPI is free; refunds are small.”

Each conclusion follows from the record in front of that person. None of them sees a customer charged twice.

A slow answer is not a failed payment

Her app
sends the payment

→

NPCI
routes it

→

Her bank
debits her account

→

Store’s bank
slow to answer

→

The till
gives up at 30 s

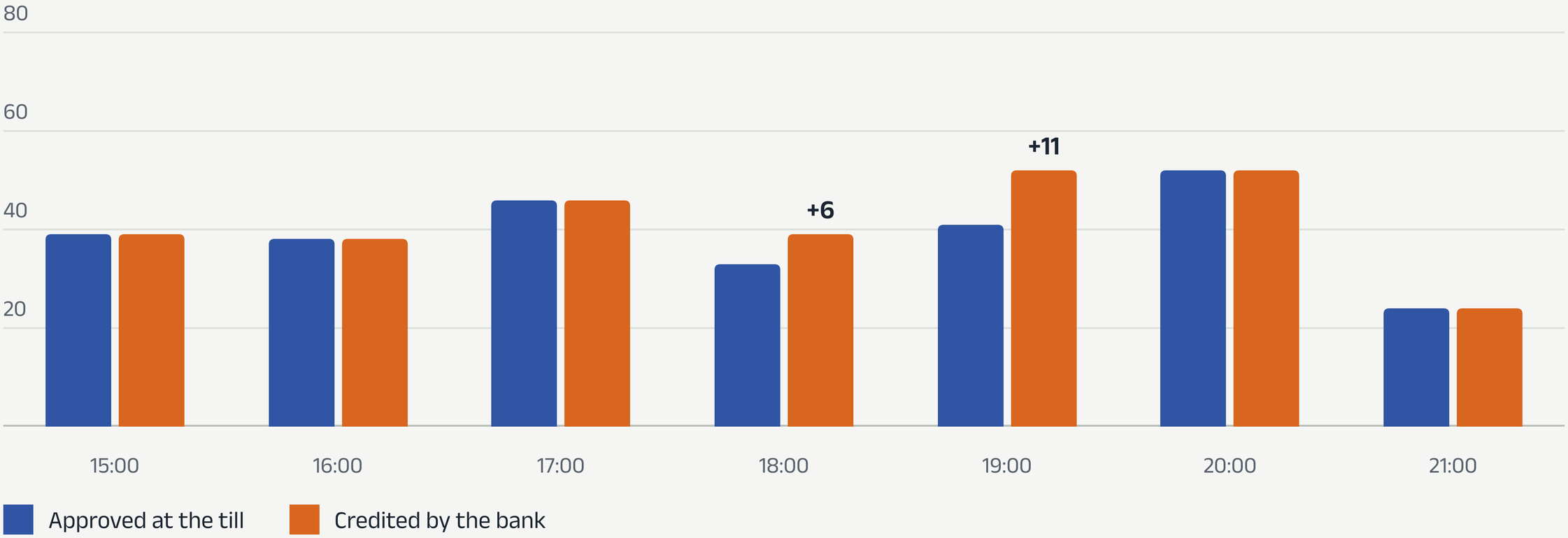
When the debit has happened and the answer is late, UPI settles it afterwards: the store’s bank confirms the credit, or the money goes back. RBI requires it back within five days of the payment, with ₹100 a day after that. The money is never lost. For a while it is in two places, and the till is the last to know.

28 of 82 UPI attempts at Store 014 got no answer within 30 seconds between 18:20 and 19:50. RBI/2019-20/67 (2019): merchant payments reversed within T+5.

Every record was right about what it saw

The till	`18:48:25 UPI ₹1,064.47 failed: no response at 18:48:55`
The till	`18:48:55 card ₹1,064.47 approved`
Provider’s report	`18:49:52 UPI 617118013344 success`
Settlement	`Sun 21 Jun: credited with Saturday’s UPI payments`
Customer support	`Tue 23 Jun, 14:00: “charged twice”`
Refund	`Sun 28 Jun, 14:00: ₹1,064.47 back`
Her money	Taken twice, for eight days

The bank was paid for more than the tills approved



UPI payments at Store 014's tills, Saturday 20 June 2026, by hour. +N: credited by the bank, marked failed at the till. Synthetic world, seed 42.

What 90 minutes cost the customers

15

customers charged twice, two of them three times: 17 extra charges, ₹7,826

4.9

median days to a refund, which came only after a complaint

All of June

Customers charged twice	24
Extra charges	26 (₹13,366)
Debited, then returned by their bank	6, within 3 days
Payment complaints	26
Still in suspense when June closed	4 (₹2,846)

Between 18:20 and 19:50 Store 014 took 119 bills; the tills gave up on 28 of 82 UPI attempts.

This is the normal case

24.07 bn

UPI transactions in September 2026, about 802 million a day

0.7–0.8%

technical declines, NPCI’s chief executive said in November 2024

90 s

before the first status check NPCI allows on a payment, since its 2025 rules

On 12 April 2025 UPI’s success rate fell to about 50% for nearly two hours, the third disruption in a week. NPCI blamed banks flooding it with status checks, and rationed them.

Sources: NPCI via Outlook Money (2 Oct 2026); Inc42 (20 Nov 2024); MediaNama on NPCI OC 215 (Apr 2025); Outlook Money (16 Apr 2025).

One name, four numbers

61%

The till

UPI attempts the till marked approved. June: 95.9%

82%

The bank

UPI attempts that ended as a credit, late ones included. June: 96.1%

100%

Bills paid

Bills the store was paid for, one way or another. June: 100.0%

87%

Charged once

Bills where no customer’s money moved twice. June: 99.9%

Payment success during the slowdown, Saturday 18:20–19:50. Over the month the four sit within a few points; in the 90 minutes that mattered they spread from 61% to 100%.

Same store, different cost: the new UPI fee

₹8,082

a month for Store 014 at June’s mix: 0.4% on 668 UPI bills above ₹2,000

From 15 October 2026: 0.4% on payments to a merchant above ₹2,000, capped at ₹300. Merchants receiving up to ₹1 lakh a month through UPI QR are exempt.

The fee on one payment

Payment	Fee
₹1,999	₹0.00
₹2,000	₹0.00
₹2,001	₹8.00
₹5,000	₹20.00
₹75,000	₹300.00

NPCI UPI and Services Steering Committee, reported by Deccan Chronicle (15 Sep 2026) and Business Standard (16 Sep 2026).

One change prevents most double charges. The other finds them all.

Wait for the status check

24 → 9

customers charged twice in June

Till time in the slowdown: 77 → 99 minutes, at the worst moment.

Bill ID on every UPI request

4.9 → 0.7

median days to a refund

Nothing left in suspense; complaints 26 → 12. Double charges still happen; every one is found the next morning.

Same customers, baskets and banks in every scenario; the changes start on 1 June. Try them in the demo.

Nothing in the records says “charged twice”

```
Till, bill B260620-110056
18:48:25 UPI failed: no response
reference: none
18:48:55 card approved

Provider’s UPI report
617118013344 18:48:25
success, confirmed 18:49:52
```

A till that hears nothing back never learns the payment’s reference number, so there is nothing to join on.

The double charge appears only when you match the provider’s successes to the till’s approvals, find a credit with no approval, and find its bill by amount and time.

Identifiers in the data are synthetic. Download June’s till payments and the provider’s report to try the match.

How to learn this domain

How UPI moves money	The parties in a payment: app, NPCI, the customer's bank, the merchant's bank, the provider. Then NPCI's 2017 circular on deemed approval.
Cards, for contrast	Glenbrook, *Payments Systems in the U.S.*: authorisation, clearing, settlement and the four-party model.
When payments fail	RBI's 2019 circular on turnaround times and compensation; NPCI's 2025 circulars on status checks and response times.
Reconciliation	The keys that join payment records: RRN, the merchant's reference, settlement dates. Credits with no sale; sales with no credit.
Practice	Download June's payment records and find every customer who paid twice.

A rough plan for a few evenings a week over about two months. Links in the post: kvinayakpai.github.io/domain-series/payments

Every record was right about what it saw. Only the customer saw both debits.

Run the scenarios and download the records:

kvinayakpai.github.io/domain-series

> **The series so far**
>
> Retail: the shelf. Logistics: the delivery. Payments: the money. One store, one month.